

NB5 — Apache Iceberg & the Catalog as Control Plane

Stack: pyiceberg + a local SQLite catalog. No JVM, no server, no cloud. Maps to slide §4 (Apache Iceberg) + §12 (Catalog = Control Plane) + deliverable bullet 5.

NB1–NB4 used Delta. Here you meet the *other* open table format — and more importantly, the thing both formats now agree is the centre of the architecture: **the catalog**.

The 2026 shift. Iceberg 1.11 moved scan planning *server-side*; Delta 4.1 shipped *catalog-managed tables*. Two rival camps, one conclusion: the catalog stopped being a name→path lookup and became the **query planner and security boundary**. That is the single biggest architectural change of the year, and it is what this notebook makes concrete.

Production equivalent: `SqlCatalog(sqlite)` ↔ `load_catalog(type="rest")` against Polaris / Unity / Lakekeeper / Glue.
Same API, same on-disk metadata — you are not learning a toy dialect.

```
In [1]: import _setup # noqa: F401 -- adds scripts/ to sys.path (file-relative)

import datetime as dtm

import pyarrow as pa

from lakehouse import catalog, count_files, du, human, namespace, reset_catalog

CAT = "nb5" # own catalog dir: NB6/NB8/`make smoke` cannot disturb it
reset_catalog(CAT) # idempotent rerun
cat = catalog(CAT)
ns = namespace(cat, "lake")
print(f"Catalog: {type(cat).__name__} namespaces: {cat.list_namespaces()}")
```

```
Catalog: SqlCatalog namespaces: [('lake',)]
```

1. Create a table *through the catalog*

Note what you do **not** do: you never pick a path. The catalog owns the layout. That indirection is exactly what lets the catalog later vend credentials, enforce row filters, and plan scans on your behalf.

`nullable=False` is not cosmetic — Iceberg tracks *required vs optional* per field, and a required field can never be silently dropped.

```
In [2]: SCHEMA = pa.schema([
    pa.field("event_id", pa.int64(), nullable=False),
    pa.field("ts", pa.timestamp("us"), nullable=False),
    pa.field("model", pa.string()),
    pa.field("latency_ms", pa.int64()),
    pa.field("cost_usd", pa.float64()),
])

tbl = cat.create_table(f"{ns}.llm_events", schema=SCHEMA)
print(f"Created {tbl.name()}")
print(f"  location: {tbl.location()}")
print(f"  metadata: {tbl.metadata_location.rsplit('/', 1)[-1]}")
print(f"  format-v{tbl.format_version}")
```

```
Created ('lake', 'llm_events')
  location: file:///home/miuu/Vin/Labs/TRACK2_DAY18_2A202601505_NguyenDuyDung/_lake
house/iceberg/nb5/warehouse/lake/llm_events
  metadata: 00000-10372e92-2d08-4939-ad3e-eb6f9d850d66.metadata.json
  format-v2
```

2. Hidden partitioning — the feature that killed Hive

In Hive you partitioned by a **derived column** (`dt=2026-08-05`) that the user had to know about and filter on by hand. Forget it, and you full-scan the table. Every data team has that outage story.

Iceberg stores the *transform* (`day(ts)`) in metadata instead. You filter on `ts` — the real column — and the engine derives the partition itself.

```
In [3]: from pyiceberg.transforms import DayTransform # noqa: E402

with tbl.update_spec() as spec:
    spec.add_field("ts", DayTransform(), "ts_day")

tbl = cat.load_table(f"{ns}.llm_events") # refresh after a metadata change
print(f"Partition spec: {tbl.spec()}")
print("\nNote: 'ts_day' is NOT a column you insert. It is derived from ts.")
```

```
Partition spec: [
  1000: ts_day: day(2)
]
```

Note: 'ts_day' is NOT a column you insert. It is derived from ts.

3. Append 10 daily batches

Each `append()` is one atomic commit → one snapshot. This is the same ACID guarantee Delta gives you, expressed through a different metadata tree.

```
In [4]: MODELS = ["claude-haiku-4-5", "claude-sonnet-4-6", "claude-opus-4-7"]
COST_PER_CALL = {"claude-haiku-4-5": 0.0004, "claude-sonnet-4-6": 0.003, "claude-op
ROWS_PER_DAY = 500
N_DAYS = 10

def day_batch(day: int) -> pa.Table:
    """One day of synthetic inference traffic."""
    models = [MODELS[i % 3] for i in range(ROWS_PER_DAY)]
    return pa.table({
        "event_id": [day * ROWS_PER_DAY + i for i in range(ROWS_PER_DAY)],
        "ts": [dtm.datetime(2026, 8, day, 3, i % 60) for i in range(ROWS_PER_DAY)],
        "model": models,
        "latency_ms": [200 + (i * 7) % 3000 for i in range(ROWS_PER_DAY)],
        "cost_usd": [COST_PER_CALL[m] for m in models],
    }, schema=SCHEMA)

for day in range(1, N_DAYS + 1):
    tbl.append(day_batch(day))

tbl = cat.load_table(f"{ns}.llm_events")
print(f"Rows: {tbl.scan().to_arrow().num_rows:,}")
print(f"Snapshots: {len(tbl.snapshots())} (one per commit)")
print(f>Data files: {tbl.inspect.files().num_rows:,}")
```

```
Rows:      5,000
Snapshots: 10 (one per commit)
Data files: 10
```

4. Scan planning — the number the catalog computes for you

`plan_files()` is the planning step itself: given a filter, which data files must actually be read? In Iceberg 1.11+ this can run **inside the catalog server**, so a laptop client never downloads the manifest tree at all.

Watch: we filter on `ts`, never on `ts_day`.

```
In [5]: scan_all = tbl.scan()
scan_one_day = tbl.scan(row_filter="ts >= '2026-08-05T00:00:00' and ts < '2026-08-06T00:00:00'")

files_all = len(list(scan_all.plan_files()))
files_one = len(list(scan_one_day.plan_files()))

print(f"Files to read, no filter: {files_all}")
print(f"Files to read, one-day filter: {files_one}")
print(f"→ Pruning ratio: {files_all / max(files_one, 1):.0f}x (target ≥ 5x)")
print(f" rows returned: {scan_one_day.to_arrow().num_rows:,}")

PRUNE_RATIO = files_all / max(files_one, 1)
assert PRUNE_RATIO >= 5, f"expected ≥5x pruning, got {PRUNE_RATIO:.1f}x"
```

```
Files to read, no filter: 10
Files to read, one-day filter: 1
→ Pruning ratio: 10× (target ≥ 5×)
rows returned: 500
```

The Hive comparison, made concrete

A Hive-style user who forgets the partition predicate reads **all** files. The Iceberg user cannot make that mistake — there is no partition column to forget. Same query intent, an order of magnitude difference in bytes scanned.

```
In [6]: # Scale the toy numbers to a realistic file size and a metered engine
# (Athena/BigQuery bill per TB scanned; $5/TB is the long-standing list price).
FILE_MB, PRICE_PER_TB, QUERIES_PER_DAY = 512, 5.0, 10_000
waste_tb = (files_all - files_one) * FILE_MB / 1_048_576

print(f"Hive user who forgets `WHERE dt=...`: reads {files_all} files")
print(f"Iceberg user filtering on `ts`: reads {files_one} files")
print(f"\nAt {FILE_MB} MB/file and ${PRICE_PER_TB:.0f}/TB scanned:")
print(f"  wasted per query: {waste_tb * 1024:.1f} GB = ${waste_tb * PRICE_PER_TB:.0f}")
print(f"  × {QUERIES_PER_DAY:,} queries/day = ${waste_tb * PRICE_PER_TB * QUERIES_PER_DAY:.0f}")
print("\nThat is the bill for one forgotten predicate. Hidden partitioning removes")
print("the opportunity to forget.")
```

```
Hive user who forgets `WHERE dt=...`: reads 10 files
Iceberg user filtering on `ts`: reads 1 files
```

```
At 512 MB/file and $5/TB scanned:
  wasted per query: 4.5 GB = $0.022
  × 10,000 queries/day = $220/day
```

That is the bill for one forgotten predicate. Hidden partitioning removes the opportunity to forget.

5. The three-tier metadata tree

This is the structure the deck draws as a chain. Now walk it for real:

```
catalog → metadata.json → manifest list → manifest files →
data files
          (schema, specs, (one per (file stats:
          snapshot ptr) snapshot) min/max, counts)
```

Every tier exists to let the tier above **skip** work below it.

```
In [7]: snaps = tbl.inspect.snapshots()
mans = tbl.inspect.manifests()
files = tbl.inspect.files()

print(f"Tier 1 metadata.json : {tbl.metadata_location.rsplit('/', 1)[-1]}")
print(f"Tier 2 manifest lists : {snaps.num_rows} (one per snapshot)")
print(f"Tier 3 manifest files : {mans.num_rows}")
```

```
print(f"          data files          : {tbl.num_rows}")
print(f"\nMetadata log entries: {tbl.inspect.metadata_log_entries().num_rows}")
print(f"Partitions tracked   : {tbl.inspect.partitions().num_rows}")
```

```
Tier 1  metadata.json      : 00011-7883f028-7830-400d-a1da-40b48482eaf5.metadata.json
Tier 2  manifest lists    : 10 (one per snapshot)
Tier 3  manifest files    : 10
        data files        : 10
```

```
Metadata log entries: 12
Partitions tracked   : 10
```

The cost of planning

Metadata is not free. Count what a planner would have to open, and compare metadata bytes to data bytes — this ratio is *why* Iceberg v4 is redesigning the metadata tree, and why server-side planning matters at scale.

```
In [8]: loc = tbl.location().replace("file://", "")
meta_bytes = du(f"{loc}/metadata")
data_bytes = du(f"{loc}/data")
print(f"data/          {human(data_bytes):>10}    ({count_files(f'{loc}/data')} parquet f
print(f"metadata/    {human(meta_bytes):>10}    ({count_files(f'{loc}/metadata', '.avro
        f'{count_files(f'{loc}/metadata', '.json')} json)")
print(f"→ metadata is {meta_bytes / max(data_bytes, 1) * 100:.1f}% of table size")
print(f"\nAt 10 rows/file this looks absurd. At 512 MB/file it is ~0.1%.")
print("Small files punish you TWICE: more data files AND more metadata to plan over
```

```
data/          47.3 KB    (10 parquet files)
metadata/      132.8 KB   (20 avro + 12 json)
→ metadata is 280.9% of table size
```

At 10 rows/file this looks absurd. At 512 MB/file it is ~0.1%.
Small files punish you TWICE: more data files AND more metadata to plan over.

6. Schema evolution by field-ID (not by name, not by position)

Parquet columns are positional. Hive matched by name. Both break under rename/reorder. Iceberg assigns every field a **permanent integer ID**; names are just labels on top of it.

So a rename is a *metadata-only* operation — zero files rewritten.

```
In [9]: from pyiceberg.types import StringType # noqa: E402

print("Field IDs before:", [(f.field_id, f.name) for f in tbl.schema().fields])

with tbl.update_schema() as upd:
    upd.add_column("tier", StringType(), doc="customer tier, added after 5000 rows")
tbl = cat.load_table(f"{ns}.llm_events")

with tbl.update_schema() as upd:
```

```

    upd.rename_column("latency_ms", "latency_millis")
tbl = cat.load_table(f"{ns}.llm_events")

print("Field IDs after :", [(f.field_id, f.name) for f in tbl.schema().fields])
print("\nlatency_ms → latency_millis kept field_id=4: a rename rewrote NO data.")
print("Old rows read back with tier=NULL – no backfill, no migration job.")

```

Field IDs before: [(1, 'event_id'), (2, 'ts'), (3, 'model'), (4, 'latency_ms'), (5, 'cost_usd')]

Field IDs after : [(1, 'event_id'), (2, 'ts'), (3, 'model'), (4, 'latency_millis'), (5, 'cost_usd'), (6, 'tier')]

latency_ms → latency_millis kept field_id=4: a rename rewrote NO data.
Old rows read back with tier=NULL – no backfill, no migration job.

```

In [10]: after = tbl.scan().to_arrow()
print(f"Rows still readable: {after.num_rows:,}")
print(f"Columns now: {after.column_names}")
print(f"tier nulls: {after.column('tier').null_count:,} (all pre-existing rows)")

```

Rows still readable: 5,000

Columns now: ['event_id', 'ts', 'model', 'latency_millis', 'cost_usd', 'tier']

tier nulls: 5,000 (all pre-existing rows)

7. Time travel by snapshot

Same idea as Delta's `versionAsOf`, different spelling. Every snapshot is addressable forever until you expire it (that's NB6).

```

In [11]: snap_ids = [s.snapshot_id for s in tbl.snapshots()]
first, last = snap_ids[0], snap_ids[-1]

print(f"snapshot[0] {first}: {tbl.scan(snapshot_id=first).to_arrow().num_rows:,} rows")
print(f"snapshot[-1] {last}: {tbl.scan(snapshot_id=last).to_arrow().num_rows:,} rows")
print(f"\nTotal snapshots retained: {len(snap_ids)}")

hist = tbl.inspect.history().to_pylist()
print("\nHistory (last 3):")
for h in hist[-3:]:
    print(f"    {h['made_current_at']} snapshot={h['snapshot_id']}")

```

snapshot[0] 2060346439476223431: 500 rows

snapshot[-1] 8408996901989835573: 5,000 rows

Total snapshots retained: 10

History (last 3):

2026-08-18 05:06:32.238000 snapshot=6918572366466123469

2026-08-18 05:06:32.260000 snapshot=3340883830715056427

2026-08-18 05:06:32.280000 snapshot=8408996901989835573

8. Partition evolution — the thing Hive genuinely cannot do

Traffic grew; daily partitions are now too coarse. In Hive this is a rewrite-the-whole-table migration. In Iceberg you change the spec and **old data stays where it is** — each data file remembers which spec wrote it.

```
In [12]: from pyiceberg.transforms import IdentityTransform # noqa: E402

with tbl.update_spec() as spec:
    spec.add_field("model", IdentityTransform(), "model_id")
tbl = cat.load_table(f"{ns}.llm_events")

tbl.append(day_batch(11).append_column("tier", pa.array(["gold"] * ROWS_PER_DAY))
          .rename_columns(["event_id", "ts", "model", "latency_millis", "cost_usd"])
tbl = cat.load_table(f"{ns}.llm_events")

specs_in_use = set(tbl.inspect.files().column("spec_id").to_pylist())
print(f"Partition specs in use across data files: {sorted(specs_in_use)}")
print(f"Total rows readable across BOTH specs: {tbl.scan().to_arrow().num_rows:,}")
print("\nTwo layouts, one table, zero rewrites. This is the feature.")
```

Partition specs in use across data files: [1, 2]

Total rows readable across BOTH specs: 5,500

Two layouts, one table, zero rewrites. This is the feature.

✓ NB5 pass criteria

Check	Target
Hidden-partition pruning ratio	$\geq 5\times$ (printed in §4)
Snapshots retained	≥ 10
Schema evolved, field IDs stable	<code>latency_millis</code> keeps <code>field_id=4</code>
Partition evolution	≥ 2 distinct <code>spec_id</code> s, table still fully readable

```
In [13]: checks = {
    "pruning ratio  $\geq 5\times$ ": PRUNE_RATIO >= 5,
    " $\geq 10$  snapshots": len(tbl.snapshots()) >= 10,
    "field_id stable on rename": [f.field_id for f in tbl.schema().fields if f.name
    " $\geq 2$  partition specs": len(specs_in_use) >= 2,
    "all rows readable": tbl.scan().to_arrow().num_rows == (N_DAYS + 1) * R
}
for k, v in checks.items():
    print(f" [{ 'PASS' if v else 'FAIL' }] {k}")
assert all(checks.values()), "NB5 incomplete — see FAIL rows above"
print("\nNB5 complete.")
```

```
[PASS] pruning ratio ≥ 5x  
[PASS] ≥ 10 snapshots  
[PASS] field_id stable on rename  
[PASS] ≥ 2 partition specs  
[PASS] all rows readable
```

NB5 complete.